Sorting factory lesson plan

22 October 2026 · Four hours · First semester with no programming or robotics background

Students work in pairs using a prepared Jupyter notebook. By the end they can change a value, predict and run a short program, describe how a sensor report affects a decision, and test a simulated sorting factory. Students do not install software or connect hardware. The optional physical demonstrations are instructor controlled; the complete learning route works offline in simulation.

Elapsed time	Duration	Activity	Evidence of learning
00:00–00:25	25 min	Introduction with eight slides	Students identify sensing, deciding and acting
00:25–00:50	25 min	A First Python instructions	Change name, repeat count and color; explain output
00:50–01:25	35 min	B Camera observations	Compare actual objects with reported labels; reject uncertainty
01:25–01:35	10 min	Break	Save notebooks
01:35–02:20	45 min	C Three rotations of 15 min	Robot sequence, stop-before-pick, camera uncertainty
02:20–03:05	45 min	D Integrated factory	Predict routes and baseline totals; change one rule
03:05–03:45	40 min	E Independent challenges	Record two predictions and experimental results
03:45–04:00	15 min	Closing	Show a change, evidence, and a simulation limitation

Exactly 240 minutes. Activity 02 robot runs inside its 15 minute rotation, not as an additional 20 minute lesson. Keep the order A then B then C then D even though the script filenames are numbered by topic.

Use the notebook for substantial explanations and the one page worksheet for navigation and records. Do not teach a list of programming definitions before students see output. The instructor guide gives prompts, pacing and solutions. Student and instructor notebooks share their executable mission code with the four scripts; regenerate with `python tools/build_notebooks.py` after editing scripts.